BANK ACCOUNT MANAGEMENT SYSTEM

Java OOP Program Documentation

1. Question

Write a Java program to create a Bank Account Management System using Object-Oriented Programming concepts.

Create a base Account class containing account number, balance, and account type. Implement operations such as displaying account details, depositing money, withdrawing money, and transferring money between accounts.

Create SavingsAccount and CurrentAccount classes using inheritance. The SavingsAccount should calculate interest, while the CurrentAccount should support an overdraft limit by overriding the withdraw() method.

Create objects for both account types and demonstrate deposit, withdrawal, interest calculation, money transfer, method overriding, inheritance, and constructor chaining using super().

2. Execution Steps

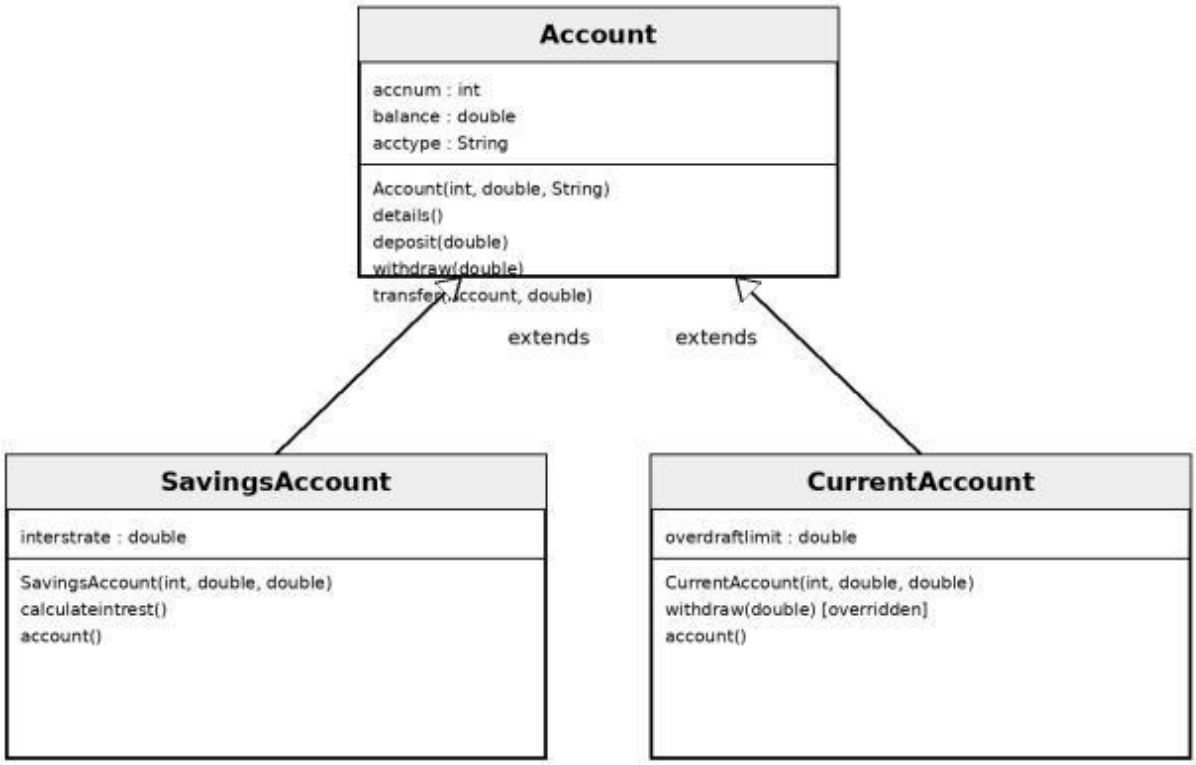
1. Open Eclipse IDE or a terminal.
2. Create a Java project/package named Excersice.
3. Create a Java class named Bank.
4. Copy the complete program into Bank.java.
5. Make sure the public class name is Bank and the file name is Bank.java.
6. Save the program.
7. Compile the program.
8. Run the Bank class containing the main() method.
9. Observe the savings account details.
10. Observe the current account details.
11. Observe deposit, withdrawal, interest, transfer, and final balance operations.
12. Verify the output in the console/terminal.

3. Algorithm

1. Start.
2. Define the Account class with account number, balance, and account type.
3. Create the Account constructor to initialize these values.
4. Define methods for displaying details, depositing, withdrawing, and transferring money.
5. Create SavingsAccount as a subclass of Account.
6. Use super() in its constructor to initialize the Account data.
7. Add an interest rate and calculate interest for the savings account.
8. Create CurrentAccount as a subclass of Account.
9. Add an overdraft limit and override withdraw() so that withdrawal can use the overdraft facility.
10. In main(), create a SavingsAccount object with account number 2000, balance 2424, and interest rate 5%.
11. Create a CurrentAccount object with account number 2154, balance 5845, and overdraft limit 500.
12. Display the initial details of both accounts.
13. Deposit 1000 into the savings account.
14. Withdraw 2000 from the current account.
15. Calculate savings-account interest and add it to the balance.
16. Transfer 500 from the savings account to the current account.
17. Withdraw 5000 from the current account using the overdraft facility.
18. Display the final account details of both SavingsAccount and CurrentAccount.
19. Stop.

4. UML Class Diagram

Account is the base class. SavingsAccount and CurrentAccount inherit from Account, and CurrentAccount overrides withdraw().



5. Java Source Code

```
package
Excersice;
class
Account{
    int accnum;
    double
    balance; String
    acctype;

    Account(int n,double
        m,String l){
        this.accnum=n;
        this.balance=m;
        this.acctype=l;
    }

    void details() {
        System.out.println("accnum
        :"+accnum);
        System.out.println("balance
        :"+balance);
        System.out.println("acctype
        :"+acctype);
    }

    void deposit(double
        amount) { if
        (amount > 0) {
            balance = balance + amount;
            System.out.println("Deposited: " +
            amount);
        }else {
            System.out.println("Invalid deposit amount");
        }
    }

    void withdraw(double amount){
        if (amount > 0 && amount <=
            balance) { balance = balance -
            amount;
            System.out.println("Withdrawn:  " + amount);
        } else {
            System.out.println(" Insufficient  balance");
        }
    }

    void transfer( Account receiver, double
        amount) { if(amount > 0 &&
        amount <= balance) {
            this.balance = this.balance - amount;
            receiver.balance = receiver.balance +
            amount; System.out.println("Transferred: " +
            amount); System.out.println("To Account: "
            + receiver.accnum);
        }else {
            System.out.println("Transfer failed");
        }
    }
}

class SavingsAccount extends
    Account{ double interstrate;

    SavingsAccount(int n,double m,double
        k){ super(n,m,"saving account");
        this.interstrate=k;
    }

    void calculateintrest() {
        double intrest = balance * (interstrate /
        100); balance = balance + intrest;
        System.out.println("value :"+intrest);
    }

    void account() {
        System.out.println("accnum :"+accnum);
        System.out.println("balance :"+balance);
        System.out.println("interstrate :"+interstrate);
```

```
    }
}

class CurrentAccount extends
    Account{ double
    overdraftlimit;

    CurrentAccount(int n,double m,double o){
        super(n,m,"current account");
        this.overdraftlimit=o;
    }

    @Override
    void withdraw(double amount){
        if (amount > 0 && amount <=
            balance+overdraftlimit) { balance =
            balance - amount;
            System.out.println("Withdrawn: " +
            amount);
        } else {
            System.out.println("exceed overdraft limt");
        }
    }

    void account() {
        System.out.println("accnum
        :"+accnum);
        System.out.println("balance
        :"+balance);
```

```
        System.out.println("overdraftlimit  :"+overdraftlimit);
    }
}

public class Bank {
    public static void main(String []args) {
        SavingsAccount s = new
        SavingsAccount(2000,2424,5); CurrentAccount c
        = new CurrentAccount(2154,5845,500);

        System.out.println("savings
        account"); s.account();

        System.out.println("\ncurrent
        account"); c.account();

        System.out.println("\nDep
        osit"); s.deposit(1000);

        System.out.println("\nwithdraw");
        c.withdraw(2000);

        System.out.println("\ninterstrate");
        s.calculateintrest();

        System.out.println("\ntransfer");
        s.transfer(c,500);

        System.out.println("\nwithdraw");
        c.withdraw(5000);

        System.out.println("\nfinal account
        details\n"); s.account();
        c.account();
    }
}
```

6. Output – Terminal Screenshot

